TECHNICAL DOCUMENTATION

🚀 Core Infrastructure & Integrations

Important core services that handle data storage, hosting, and versioning. These are the primary integration points for the system.

MongoDB
Primary Database

Cloudinary
Media & File Storage

Vercel
Cloud Hosting Platform

GitHub
Source Control & CI/CD

🏗️ Core Architecture

The project follows a modern **Decoupled MERN Architecture** using Next.js for the interface and Node.js fc processing engine. This ensures dedicated resources for both frontend interactions and backend logic.

🎨 Frontend Technologies

Framework

Next.js 15 & React 19

Utilizing Server Components and modern hooks for ultra-fast performance.

Styling

Tailwind CSS 4.0

Premium responsive design with a custom-bu utility system.

Analysis

Scheduling

Chart.js & Recharts

Dynamic data visualization for resident and staff analytics.

FullCalendar & Big Calendar

Enterprise-grade scheduling for managemen tasks.

⚙️ Backend & API

Engine

Node.js & Express

Secure RESTful API handling all business logic and data routing.

Security

JWT & Bcrypt

Military-grade encryption for passwords and secure token-based sessions.

Communication

Nodemailer

Integrated email system for automated notifications.

Handling

Multer & Cloudinary

Professional image processing and cloud sto synchronization.

🗄️ Database & Cloud Details

- **Database System:** MongoDB Atlas (Cloud-based NoSQL database).
- **Media Management:** Cloudinary CDN (Automated image optimization and hosting).
- **Version Control:** GitHub Repository (Master branch deployment).
- **Production Hosting:** Vercel Edge Network (High-availability serverless hosting).

🚀 Deployment Status

The system is fully integrated for modern dev-ops workflow:

- **Source Control:** GitHub (Master branch).
- **Deployment Platform:** Vercel (Production environment).

- **Pipeline:** Automated CI/CD (Github to Vercel).